

EAGLE, AK, USA

WMO: 701975

Lat: 64.777N

Lon: 141.148W

Elev: 277

StdP: 98.04

Time Zone: -9.00 (NAK)

Period: 98-19

WBAN: 26422

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -45.3	(c) -41.9	(d) -37.5	(e) 0.1	(f) -33.1	(g) -36.4	(h) 0.1	(i) -32.4	(j) 9.0	(k) -23.6	(l) 8.3	(m) -21.9	(n) 0.2	(o) 180	(p) 0.682

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 14.5	(c) 27.5	(d) 16.1	(e) 25.7	(f) 15.1	(g) 23.8	(h) 14.3	(i) 17.4	(j) 24.6	(k) 16.3	(l) 23.3	(m) 15.4	(n) 21.7	(o) 2.5	(p) 20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.0	11.0	18.4	13.6	10.1	17.7	12.5	9.3	17.2	49.9	24.6	46.7	23.2	43.8	21.6	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.3	5.9	5.2	-46.0	30.3	3.3	1.4	-48.4	31.3	-50.3	32.2	-52.1	33.0	-54.5	34.0
			-36.3	19.3	1.6	2.5	-37.4	21.1	-38.3	22.6	-39.2	24.0	-40.3	25.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.9	-21.0	-17.0	-13.4	-1.5	7.8	13.7	15.1	11.6	5.9	-3.4	-15.1	-18.6
	DBStd	14.86	11.37	10.22	8.72	6.35	4.37	3.02	2.51	3.49	3.86	5.79	8.91	9.77
	HDD10.0	5094	959	755	724	346	96	4	2	23	129	417	753	887
	HDD18.3	7762	1218	988	982	596	326	141	104	211	373	675	1003	1145
	CDD10.0	381	0	0	0	0	28	116	158	72	7	0	0	0
	CDD18.3	6	0	0	0	0	0	2	3	1	0	0	0	0
	CDH23.3	469	0	0	0	0	46	167	179	76	0	0	0	0
	CDH26.7	80	0	0	0	0	6	28	30	16	0	0	0	0
Wind	WSAvg	1.6	1.8	1.7	1.3	1.6	1.8	1.6	1.3	1.2	1.4	1.4	1.7	1.8
Precipitation	PrecAvg	296	13	11	8	9	23	46	57	49	30	25	18	17
	PrecMax	399	42	22	39	21	48	87	123	98	84	48	57	50
	PrecMin	171	3	0	0	0	0	9	16	15	0	3	0	1
	PrecStd	65	9	6	8	6	14	20	26	23	18	11	13	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.1	6.7	9.0	18.5	27.6	29.3	29.3	28.8	20.6	11.7	5.3	5.3
		MCWB	2.4	2.0	3.8	10.2	13.6	16.9	16.8	16.7	12.1	6.2	2.1	1.8
	2%	DB	1.6	2.9	6.5	14.2	24.3	27.2	27.3	25.6	18.0	8.9	1.3	2.0
		MCWB	-0.6	-0.5	1.7	6.3	12.6	16.0	16.2	16.4	10.6	4.4	-0.7	-0.5
	5%	DB	-2.8	0.6	3.5	11.9	22.1	25.2	25.4	22.6	16.4	7.2	-0.9	-2.1
		MCWB	-4.5	-1.8	-0.3	5.0	11.7	14.5	15.5	14.8	9.9	3.5	-2.5	-3.8
	10%	DB	-6.5	-2.6	0.6	9.6	19.0	23.3	23.4	20.1	14.6	4.9	-2.9	-5.3
		MCWB	-7.6	-4.6	-2.2	3.8	10.2	13.5	14.9	13.8	9.2	2.2	-4.3	-6.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	2.3	4.6	10.0	15.8	21.6	18.7	19.8	12.8	7.0	2.6	2.1
		MCDB	5.9	6.4	8.7	17.6	23.0	25.9	26.2	23.0	18.3	10.4	5.4	5.5
	2%	WB	-0.2	0.0	2.2	7.0	13.5	17.1	17.2	16.9	11.6	5.0	-0.4	-0.4
		MCDB	2.4	2.9	6.2	13.9	22.1	24.8	24.3	24.1	16.5	8.2	1.2	2.0
	5%	WB	-3.4	-1.7	0.1	5.3	12.2	15.5	16.4	15.5	10.6	3.6	-2.3	-3.4
		MCDB	-1.9	0.8	3.5	11.2	20.4	22.6	23.2	21.5	15.2	6.7	-0.7	-1.6
	10%	WB	-6.9	-4.2	-2.2	4.0	10.7	14.3	15.6	14.2	9.7	2.4	-4.2	-6.0
		MCDB	-5.7	-2.3	0.6	9.1	18.4	21.5	22.0	19.1	14.3	5.0	-2.9	-4.7
Mean Daily Temperature Range	5% DB	MDBR	8.7	11.7	16.1	15.8	16.4	16.2	14.5	14.3	13.1	9.0	8.1	8.4
		MCDBR	10.6	13.2	16.7	18.1	22.1	20.4	19.7	19.7	17.4	11.2	8.5	9.5
		MCWBR	9.1	10.4	12.4	11.1	11.4	10.5	10.0	11.1	10.8	7.6	7.1	7.9
	5% WB	MCDBR	10.8	12.8	15.4	17.3	20.2	17.3	16.4	17.1	15.3	10.2	8.3	9.5
		MCWBR	9.3	10.2	11.4	11.0	11.2	9.5	9.0	10.1	9.9	7.3	7.1	7.9
Clear-Sky Solar Irradiance	taub		0.189	0.245	0.254	0.326	0.354	0.384	0.400	0.370	0.318	0.263	0.171	0.174
	taud		1.915	2.199	2.284	2.174	2.370	2.310	2.292	2.391	2.508	2.414	1.955	1.931
	Ebn at Noon		339	674	848	839	853	828	799	792	772	661	368	122
	Edh at Noon		24	54	78	112	103	112	111	92	66	44	21	8
All-Sky Solar Radiation	RadAvg		0.15	0.85	2.68	4.77	5.40	5.57	4.68	3.71	2.31	0.98	0.24	0.05
	RadStd		0.01	0.06	0.15	0.28	0.34	0.49	0.36	0.34	0.21	0.12	0.02	0.00

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	+2.75	+3.17	N/A	N/A	N/A	N/A	N/A	+55	N/A

Nomenclature: See separate page